MESIN KONVERSI ENERGI

A5 • Rumus inti dengan notasi matematika • Gunakan satuan konsisten

01 PISTON & BAHAN BAKAR

$$V_{d,tot} = z\frac{\pi D^2}{4}L, \quad r = \frac{V_{d,sil} + V_c}{V_c}$$

$$P_i = \text{IMEP}\, V_{d,tot}\frac{N}{60a}$$

$$P_b = \frac{2\pi N\tau}{60}, \quad P_f = P_i - P_b$$

$$\eta_m = \frac{P_b}{P_i}, \quad \eta_b = \frac{P_b}{\dot{m}_f \text{LHV}}$$

$$\text{BSFC} = \frac{3.6\times10^6}{\eta_b\,\text{LHV}}\ [\text{g/kWh}]$$

a=2: 4 langkah; a=1: 2 langkah; N: rpm.
LHV: kJ/kg. MEP: kPa + m³ → daya kW.

02 DASAR & PROSES IDEAL

$$Q - W = \Delta U, \quad pv = RT$$

$$\Delta u = c_v\Delta T, \quad \Delta h = c_p\Delta T$$

$$k = \frac{c_p}{c_v}, \quad R = c_p - c_v$$

$$pv^k = C, \quad \frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}$$

$$v = \text{konstan}: \ w = 0, \ q = c_v\Delta T$$

$$p = \text{konstan}: \ w = R\Delta T, \ q = c_p\Delta T$$

$$q = 0: \ w = c_v(T_{awal} - T_{akhir})$$

03 OTTO & DIESEL

$$r = \frac{V_1}{V_2}, \quad \rho = \frac{V_3}{V_2}, \quad T_2 = T_1 r^{k-1}$$

$$\text{Otto}: \quad \eta_O = 1 - \frac{1}{r^{k-1}}$$

$$q_{in} = c_v(T_3 - T_2), \quad T_4 = \frac{T_3}{r^{k-1}}$$

$$\text{Diesel}: \ \eta_D = 1 - \frac{\rho^k - 1}{k(\rho-1)r^{k-1}}$$

$$q_{in} = c_p(T_3 - T_2), \quad T_3 = \rho T_2$$

$$T_4 = T_3\left(\frac{\rho}{r}\right)^{k-1}$$

$$q_{out} = c_v(T_4 - T_1), \quad w_{net} = q_{in} - q_{out}$$

04 BRAYTON & BOILER

$$w_c = h_2 - h_1, \quad w_t = h_3 - h_4$$

$$q_{in} = h_3 - h_2, \quad w_{net} = w_t - w_c$$

$$\eta_{th} = \frac{w_{net}}{q_{in}}, \quad \text{BWR} = \frac{w_c}{w_t}$$

$$\dot{m} = \frac{p_1\dot{V}_1}{RT_1}, \quad P_{net} = \dot{m}\, w_{net}$$

$$\dot{Q}_s = \dot{m}_s(h_{out} - h_{in})$$

$$\dot{m}_f = \frac{\dot{Q}_s}{\eta_{boiler}\,\text{LHV}}$$

$$\text{HR} = \frac{3600}{\eta_{fuel\to elec}}\ [\text{kJ/kWh}]$$

05 RANKINE / ECE

1 → pompa → 2 → boiler → 3 → turbin → 4 → kondensor → 1; ideal: $s_2{=}s_1$, $s_4{=}s_3$.

$$w_p = h_2 - h_1 \simeq v_1(p_2 - p_1)$$

$$w_t = h_3 - h_4, \quad q_{in} = h_3 - h_2$$

$$\eta_{th} = \frac{w_t - w_p}{q_{in}}, \quad P_{net} = \dot{m}(w_t - w_p)$$

$$x = \frac{s - s_f}{s_{fg}}, \quad h = h_f + xh_{fg}$$

Perbaikan: superheat, p kondensor turun, reheat, regenerasi, optimasi p boiler.

06 AKTUAL & KONSEP PENTING

$$\eta_t = \frac{h_{in} - h_{out,a}}{h_{in} - h_{out,s}}$$

$$\eta_c = \frac{h_{out,s} - h_{in}}{h_{out,a} - h_{in}}$$

$$\eta_p = \frac{h_{out,s} - h_{in}}{h_{out,a} - h_{in}}$$

$$\eta_b = \eta_m\eta_i = \eta_m\eta_{comb}\eta_{conv}$$

a: aktual; s: isentropik. ηcomb ≠ ηi.
ICE: pembakaran internal; ECE: eksternal.
HHV > LHV: HHV mencakup kondensasi air.
SI knock: end-gas menyala sendiri.
CI knock: ignition delay → p naik cepat.

SATUAN & ASUMSI

1 bar=100 kPa; 1 cc=10^{-6} m³; 1 kWh=3600 kJ; 1 hp≈0,7457 kW; T[K]=T[°C]+273,15.
ηfuel→elec = ηboiler·ηcycle·ηgen jika ηcycle berbasis kalor ke fluida; jangan hitung boiler dua kali.
Otto/Diesel: gas ideal, kalor jenis konstan; 1–2 & 3–4 isentropik. Uap: gunakan tabel.
Q masuk (+), W keluar (+); q,w per kg; ηi,ηb berbasis fuel. Cetak A5 pada 100% / Actual Size.